Xu Duan

Male | Jul. 18, 2002
(737) 341-7377 | xuduan@utexas.edu

EDUCATION

The University of Texas at Austin **Aug. 2024 - May. 2026 (Expected)**

- Master Student, Mechanical Engineering, Dynamic System and Control Area
- Co-Advised by Dr. van Oort and Dr. Chen

Shanghai Jiao Tong University **Sep. 2019 - Jul. 2023**

- Bachelor of ***Engineering*** in Naval Architecture and Marine Engineering (Major ranked #1 globally)
- GPA: **3.79/4.0**, **89/100**

Highlighted Coursework: *Introduction to Modern Control, Digital Control, Finite Element Method, Computational Fluid Dynamics, Dynamics of Mechanical System, Convex Optimization, Thinking and Approach of Programming (C++), Data Structures (C++), Engineering Analysis, etc.*

HONORS & AWARDS

Philip C. and Linda L. Lewis Foundation Graduate Fellowship	Aug. 2024
CCS (China ship Classification Society) Scholarship	Sep. 2022
Shengshen Scholarship	Sep. 2021
CCS (China ship Classification Society) Scholarship	Sep. 2020

INTERNSHIPS

Haipeng Marine Engineering (Shanghai) Co. **Jul. 01, 2023 - Aug. 31, 2023**

Engineering Intern

- Marine environmental assessment, including data collection and analysis of marine environment;
- Performed hydrodynamic and structural analysis of marine structures, including marine platforms and marine pipelines, with HydroStar and Abaqus to calculate the hydrodynamic load and motion.

Bureau Veritas S.A. at Shanghai **Jul. 01, 2022 - Aug. 31, 2022**

Engineering Intern

- Inspected the technical drawings of ships and marine platforms to make sure that these structures would conform to BV rules.

PROJECTS

Control System Design for an Underwater Robot

- Committed to establish mathematical model and simulation of 6-DOF underwater robot with Simulink;
- Performed the design of the Proportional Integral Differential (PID) control system according to the deviations of the actual position of the robot from the target position.

Design and Calculation of the Driving Gear for an Automobile

- Committed to the simulation of the gear's motion and the verification of the gear's strength;
- Judged the structural strength to select the steel materials based on simulation results, calculated using the Finite Element Method (FEM), simulated based on Abaqus.

Book Lamp Control System using PLC

- Designed and implemented a control system for opening and closing book lamps using Programmable Logic Controllers (PLC);
- Developed and debugged program to process analog signals from a photosensitive resistor, converting them into digital signals to monitor and adjust light intensity.

Wind Speed Prediction with Neural Networks

- Committed to perform short-term forecasting of future wind speeds using an LSTM (Long Short-Term Memory) neural network based on the original wind speed data;
- Employed some signal processing methods to preprocess the raw wind speed data.

OTHER INFORMATION

Languages: Fluent in English (IELTS 7.0, GRE 336/4.0), Native in Chinese (Simplified).
Programming: Proficient in Python, C/C++, Linux, MATLAB, Fortran
Software: Abaqus, Solidworks, Rhino, AutoCad, Simulink, Microsoft Office suite
Transferable Attributes:

- Superior diagnostic skills for issues and innovative invention practices;
- Capacity to coordinate and optimize solutions and conduct flexible commissioning;
- Collaborative teamwork skills;
- Foresight and vision to evaluate the complexity of projects.